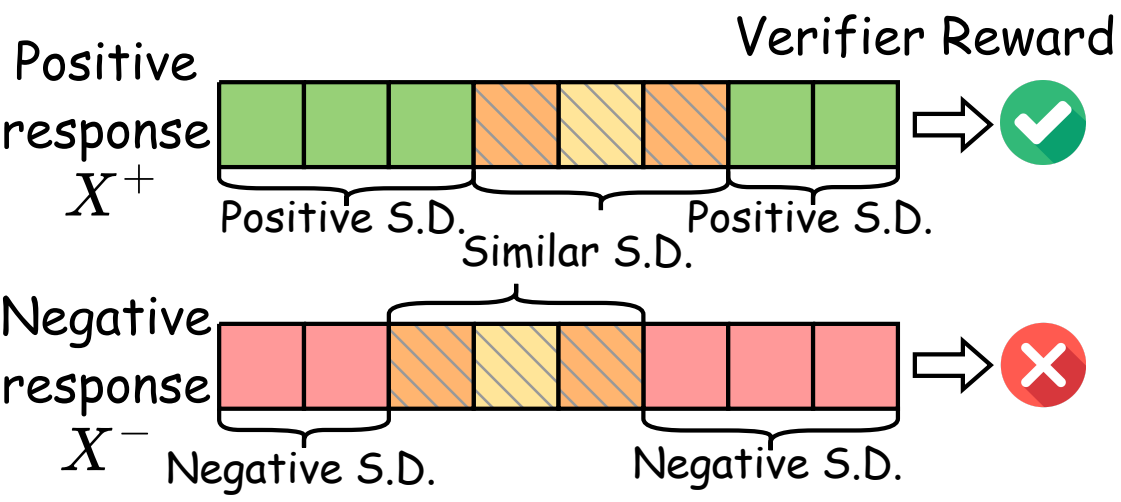

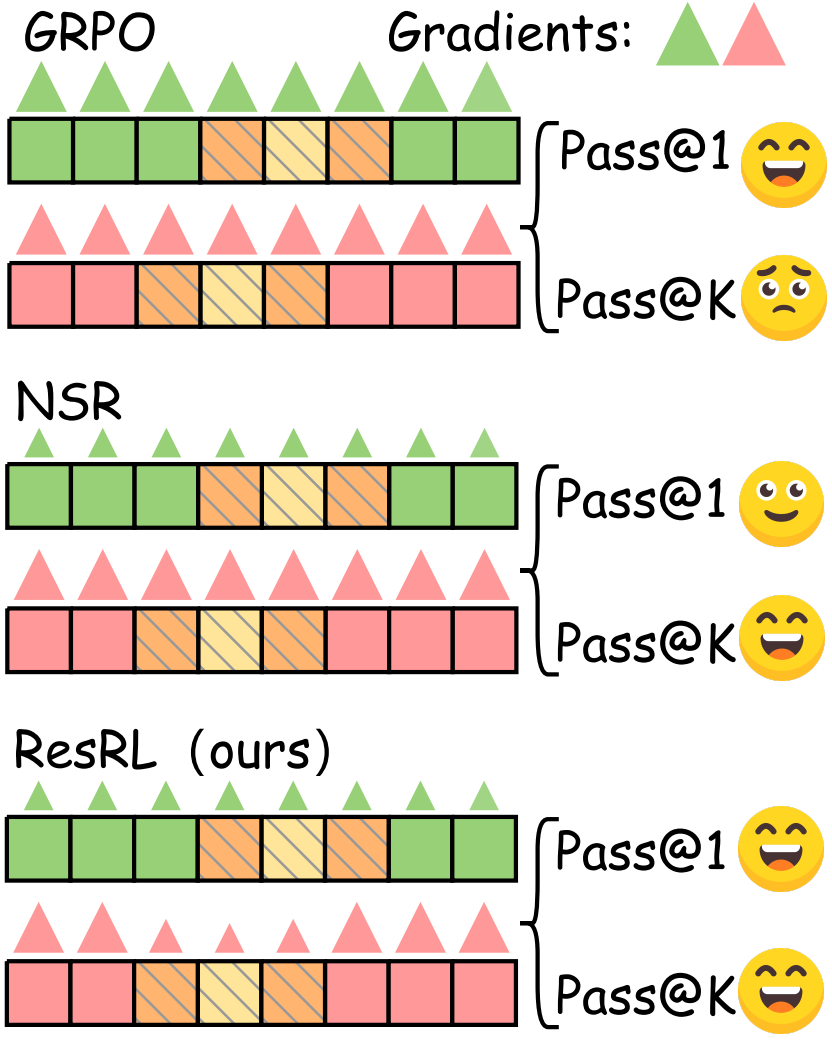

Negative Sample Projection Residual

SVD Decomposition: $X^+ = U\Sigma V^\top, P_{\mathcal{S}} = V_k V_k^\top$

Projection Residual: $R_{i,t} = \frac{1}{d}\|X^- - P_{\mathcal{S}} X^-\|_2^2$

Gradient Weight: $\tilde{A}_{i,t} = \Big(\xi + (1-\xi)\,\mathrm{f}(R_{i,t})\Big)\hat{A}_i$